

ASSESSMENT – 2

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Course Code :DSA0503

Course Name :Query Processing for Data Science

Assessment :CO2-AT1

1. Examination Result Management System (10 Marks)

A college maintains student details and examination results in PostgreSQL. Design Student and Result tables using appropriate primary and foreign keys. Write a Python program to create the tables, insert sample records, calculate the total marks of each student using SQL queries, and display the student names along with their total marks.

Aim: To create Student and Result tables in PostgreSQL, insert sample records, calculate the total marks of each student using SQL queries, and display the student names along with their total marks using Python.

Python Code:

```
import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college",
    user="postgres",
    password="pawan1611"
)

cur = conn.cursor()

cur.execute("""
CREATE TABLE IF NOT EXISTS Student(
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50)
)
""")

cur.execute("""
CREATE TABLE IF NOT EXISTS Result(
    ResultID INT PRIMARY KEY,
    StudentID INT,
    Subject1 INT,
    Subject2 INT,
    Subject3 INT,
    FOREIGN KEY(StudentID) REFERENCES Student(StudentID)
)
""")
```

```
cur.execute("INSERT INTO Student VALUES (1,'Alice') ON CONFLICT DO NOTHING")
cur.execute("INSERT INTO Student VALUES (2,'Bob') ON CONFLICT DO NOTHING")
```

```
cur.execute("INSERT INTO Result VALUES (101,1,85,90,88) ON CONFLICT DO
NOTHING")
cur.execute("INSERT INTO Result VALUES (102,2,78,82,80) ON CONFLICT DO
NOTHING")
```

```
conn.commit()
```

```
cur.execute("""
SELECT Student.StudentName,
       (Result.Subject1 + Result.Subject2 + Result.Subject3) AS TotalMarks
FROM Student
JOIN Result
ON Student.StudentID = Result.StudentID
""")
```

```
rows = cur.fetchall()
```

```
print("Student Name\tTotal Marks")
for row in rows:
    print(row[0], "\t\t", row[1])
```

```
cur.close()
conn.close()
```

Sample Output:

Student Name	Total Marks
Alice	263
Bob	240

Result

The program successfully creates the **Student** and **Result** tables, inserts sample records, calculates the total marks of each student using SQL queries, and displays the student names along with their total marks.

2. Vehicle Service Management System (10 Marks)

A vehicle service center wants to maintain customer and vehicle service information using MySQL. Design Customer and Service tables with appropriate relationships. Write a Python program to create the tables, insert service records, update the service status of a vehicle, and retrieve customer details along with their service history.

Aim

To create Customer and Service tables in MySQL, insert service records, update the service status of a vehicle, and retrieve customer details along with their service history using Python.

Python Code:

```
import mysql.connector
```

```
conn = mysql.connector.connect(  
    host="localhost",  
    user="root",  
    password="pawan1611",  
    database="service_db"  
)
```

```
cur = conn.cursor()
```

```
cur.execute("""  
CREATE TABLE IF NOT EXISTS Customer(  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(50),  
    VehicleNo VARCHAR(20)  
)  
""")
```

```
cur.execute("""  
CREATE TABLE IF NOT EXISTS Service(  
    ServiceID INT PRIMARY KEY,  
    CustomerID INT,  
    ServiceType VARCHAR(50),  
    ServiceStatus VARCHAR(20),  
    FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)  
)  
""")
```

```

""")

cur.execute("INSERT IGNORE INTO Customer VALUES (1,'Arun','TN10AB1234')")
cur.execute("INSERT IGNORE INTO Customer VALUES (2,'Priya','TN20CD5678')")

cur.execute("INSERT IGNORE INTO Service VALUES (101,1,'General Service','Pending')")
cur.execute("INSERT IGNORE INTO Service VALUES (102,2,'Oil Change','Pending')")

conn.commit()

cur.execute("UPDATE Service SET ServiceStatus='Completed' WHERE ServiceID=101")

conn.commit()
cur.execute("""
SELECT Customer.CustomerName,
       Customer.VehicleNo,
       Service.ServiceType,
       Service.ServiceStatus
FROM Customer
JOIN Service
ON Customer.CustomerID = Service.CustomerID
""")

rows = cur.fetchall()

print("Customer Details and Service History")
for row in rows:
    print(row)

cur.close()
conn.close()

```

Sample Output:

```

Customer Details and Service History
('Arun', 'TN10AB1234', 'General Service', 'Completed')
('Priya', 'TN20CD5678', 'Oil Change', 'Pending')

```

Result

The program successfully creates the **Customer** and **Service** tables, inserts service records, updates the service status of a vehicle, and retrieves customer details along with their service history.